

## Sample proofs – Asymptotic Analysis

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1. Prove  $n^2 + 3n + 2 = \Theta(n^2)$

### Lower Bound

By definition, we need to find positive constants  $c_1, c_2, n_0$  such that

$$0 \leq c_1 n^2 \leq n^2 + 3n + 2 \leq c_2 n^2$$

Lets assume,  $c_1 = 1, n_0 = 1$ .

For  $n \geq 1$ ,

$$n^2 \leq n^2 + 3n + 2$$

because  $3n + 2 \geq 0$ .

Therefore,

$$\boxed{n^2 \leq n^2 + 3n + 2}$$

### Upper bound

Now we need some  $c_2$  such that

$$n^2 + 3n + 2 \leq c_2 n^2.$$

Since  $n \geq 1$ ,

$$n \leq n^2$$

and

$$1 \leq n^2.$$

Therefore,

$$n^2 + 3n + 2 \leq n^2 + 3n^2 + 2n^2 = 6n^2.$$

Thus we can choose  $\boxed{c_2 = 6}$ .

Hence, for every  $n \geq 1$ ,

$$n^2 \leq n^2 + 3n + 2 \leq 6n^2.$$

Therefore,

$$\boxed{n^2 + 3n + 2 = \Theta(n^2)}$$

with one valid choice being

$$\boxed{c_1 = 1, c_2 = 6, n_0 = 1}.$$

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2. Prove  $2n + 3 = O(n)$

We need to find  $c > 0$  and  $n_0$  such that

$$2n + 3 \leq cn$$

for all  $n \geq n_0$ , let  $n_0 = 1$  then,

$$1 \leq n.$$

Multiplying by 3,

$$3 \leq 3n.$$

Therefore,

$$2n + 3 \leq 2n + 3n = 5n.$$

So we can choose

$$\boxed{c = 5, n_0 = 1}.$$

Therefore,

$$\boxed{2n + 3 = O(n)}.$$

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3. Prove  $4n^2 + 2n + 8 = O(n^2)$

We need

$$4n^2 + 2n + 8 \leq cn^2.$$

For  $n \geq 1$ ,

$$n \leq n^2$$

and

$$1 \leq n^2.$$

Therefore,

$$2n \leq 2n^2$$

and

$$8 \leq 8n^2.$$

Thus, adding  $4n^2$  on both sides, we get

$$4n^2 + 2n + 8 \leq 4n^2 + 2n^2 + 8n^2 = 14n^2.$$

Therefore, we can choose

$$\boxed{c = 14, n_0 = 1}.$$

Hence,

$$\boxed{4n^2 + 2n + 8 = O(n^2)}.$$

4. Prove  $n^2 + 5n = \Omega(n^2)$

We need to find  $c > 0, n_0$  such that

$$cn^2 \leq n^2 + 5n$$

For  $n \geq n_0$ , Since

$$5n \geq 0$$

for  $n \geq 0$ ,

$$n^2 + 5n \geq n^2.$$

Therefore,

$$1 \cdot n^2 \leq n^2 + 5n.$$

So we can choose

$$\boxed{c = 1, n_0 = 1}.$$

Hence,

$$\boxed{n^2 + 5n = \Omega(n^2)}.$$

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5. Prove  $6n^3 + n = \Omega(n^3)$

Again, we need

$$cn^3 \leq 6n^3 + n.$$

Since  $n \geq 0$ , Therefore,

$$6n^3 + n \geq 6n^3.$$

So we can choose

$$c = 6, n_0 = 1.$$

Thus,

$$6n^3 + n = \Omega(n^3).$$

We could also choose  $c = 1$ , because

$$n^3 \leq 6n^3 + n.$$

So  $c = 6$  isn't the only possible answer.

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For all these questions, you just need to find a pair of values  $c, n_0$  ( $c_1, c_2, n_0$  for tight bounds) that satisfy the relationship you have been asked to prove. There might be multiple pairs of values. You just need to show one to prove. If you can't find any values of  $c$  and  $n_0$  that satisfies the relationship then state that in your answer – this type of question can also be asked as True/False.